

GREYWATER MANAGEMENT

District-Level Toolkit

Rural areas • Planning • Treatment • Reuse • Monitoring

Source

A toolkit for district-level officials on Greywater Management



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officials on Greywater
Management**

1. Purpose and Structure of the Toolkit

- Provides step-by-step guidance for district, block and Gram Panchayat (GP) functionaries.
- Covers planning, implementation, operation & maintenance, and monitoring of greywater management.
- Sets out connected sub-activities, responsibilities, key specifications and practical implementation tools.

Background & need

Brief on rural
greywater
management

Guiding principles

Technology options
+ SBM (G) Phase-II

Step-by-step rollout

Planning and
implementation

Supportive tools

Implementation tools /
annexures

2. Guiding Principles and Treatment Approach

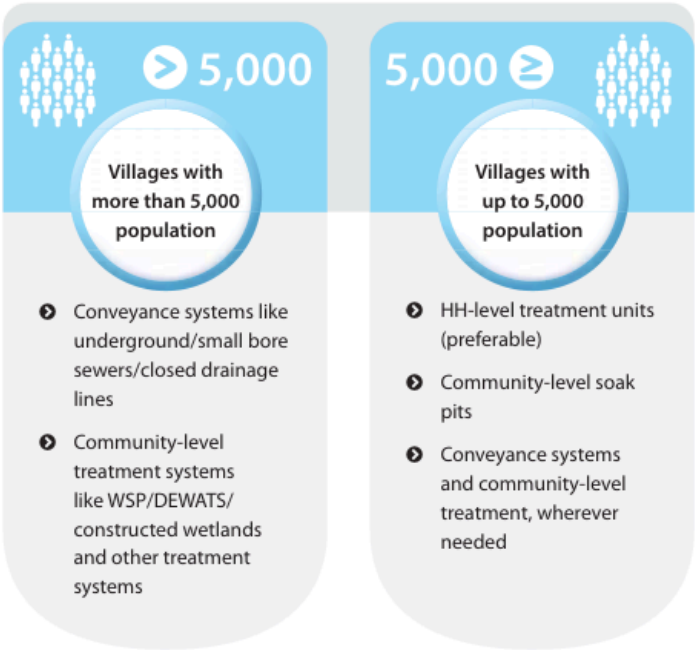
- SBM (G) Phase-II identifies effective liquid-waste management as a key objective.
- Treatment should be planned as close as possible to the point of greywater generation.
- For villages above 5,000 population: conveyance systems plus community treatment such as WSP, DEWATS or constructed wetlands.
- For villages up to 5,000 population: household-level units are preferred; community systems are used where household units are not feasible.

Guiding Principles of SLWM: SBM (G) Phase-II

The central government released the guidelines for Swachh Bharat Mission (Grameen) Phase-II in February, 2020. According to the guidelines, effective management of liquid waste by at least 80 per cent of households is one of the key objectives of this phase.

2.1 Greywater Treatment

Figure 1: Treatment Options for Greywater



3. Greywater Management Technologies

Household level

Soak pit • Leach pit • Magic pit • Kitchen garden

Small group

5–15 household kitchen garden / leach pit

Community level

WSP / Constructed Wetland / DEWATS / SBT / Phytorid

Conveyance

Small-bore sewers / closed drainage lines

Select technology based on space, soil, water table and local conditions

4. Technology Selection by Site Condition

- Permanent high-water-table / waterlogged areas:
 - Kitchen garden where household space permits.
 - Community leach pit where suitable space and conditions exist.
 - Where space is inadequate, use suitable small-bore / WSP / reed-bed / DEWATS / SBT systems.
- Hard or rocky strata:
 - Kitchen garden where space is available.
 - Community leach pit where permeable soil is available nearby.
 - Where nearby space is unavailable, use suitable conveyance and community treatment systems.

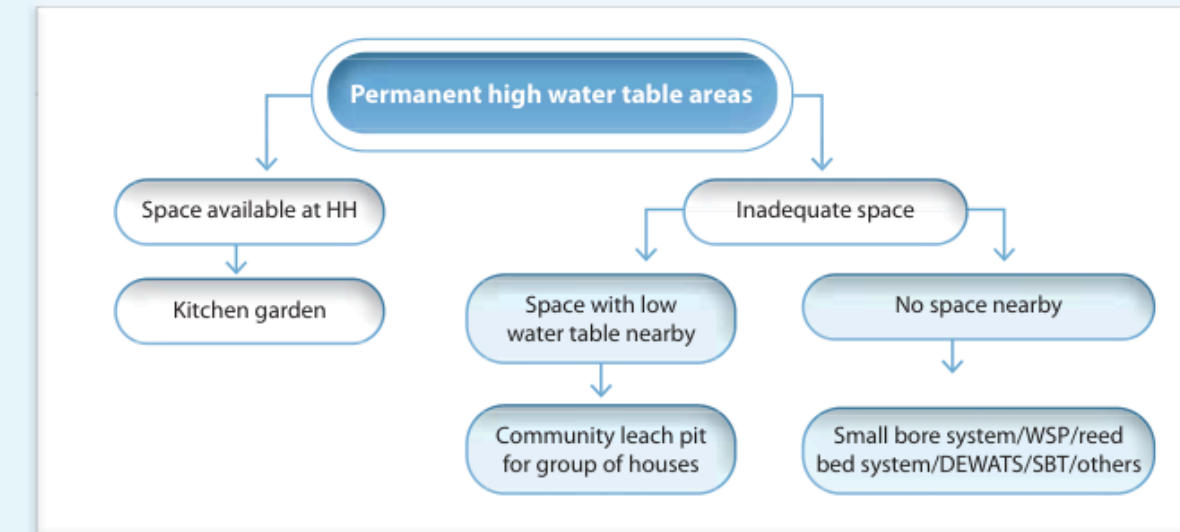
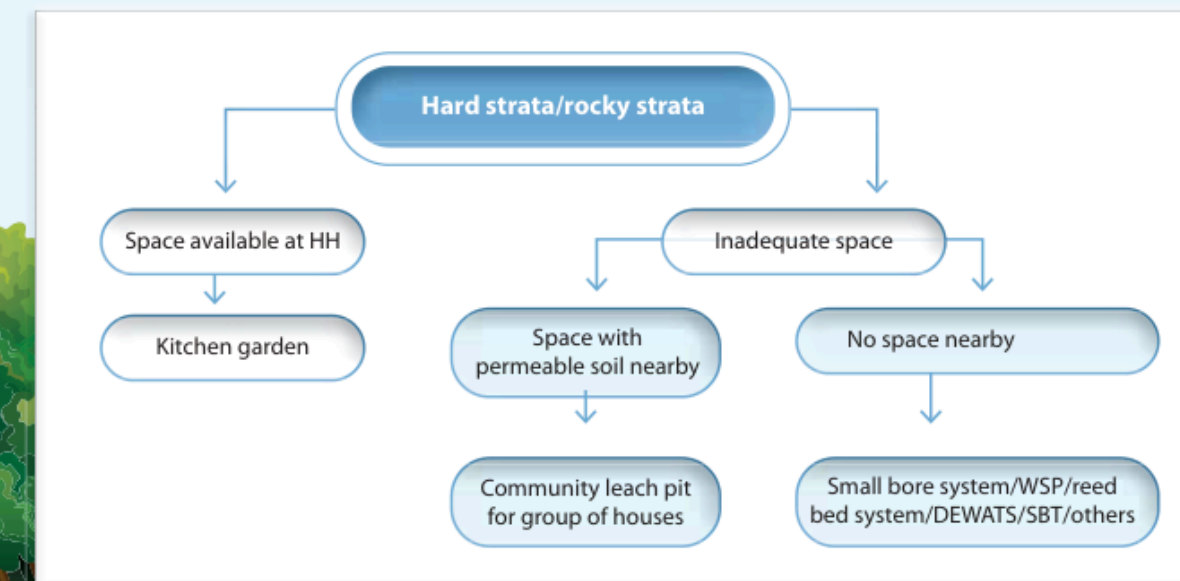


Figure 4: Technology Options for Hard Strata Areas



5. Village-Level Planning for Greywater Management

- Prepare a Village Action / Swachhata Plan (VAP/VSP).
- Record existing household and community treatment units and conveyance systems.
- Identify households needing household-level or community-level connections.
- Assess greywater quality, land availability, conveyance needs and estimated greywater generation.
- Record reuse/recharge arrangements and funds received and spent.
- Plan treatment capacity, connected households and additional conveyance required.

The Village Action/Swachhata Plan (VAP/VSP) should cover the following aspects of greywater management:

- Existing number of households connected to household-level treatment units
- Details of existing conveyance systems, if any
- Number and details of community-level greywater treatment units
- Number of households that need to be connected to household-level treatment units
- Number of households that need to be connected to community-level treatment units
- Quality of the greywater generated so that community treatment facilities may be planned
- Availability of land for development of community treatment facilities
- Requirements for conveyance systems
- Estimated amount of greywater generated
- Details of greywater reuse and recharge, if any
- Funds received for greywater management and expenditure

Table 2: Checklist for GP-level implementation for Greywater Management

A	Situation analysis	
1	Whether all HHs are connected to HH/community-level greywater treatment units?	(Yes/No)
2	If no, what is the number of HHs that are not connected to HH/community-level greywater treatment units?	_____
3	How many HHs have HH-level greywater treatment units?	Soak pits_____ Leach pits_____ Magic pits_____ Kitchen gardens _____
4	How many community-level greywater treatment units are there in the GP?	

6. Roles and Responsibilities

District

- Support GP planning and implementation
- Finalise targets and financial plans
- Provide technical support and capacity building
- Conduct IEC activities and reviews
- Support O&M and monitoring
- Maintain district-level records

Block

- Orient and coordinate GPs
- Promote parallel implementation
- Support convergent action
- Undertake IEC at block/village level
- Prepare block-level plans
- Monitor progress and promote reduce–reuse–recharge

GPs / Households

- Use fresh water judiciously
- Set up household treatment units where feasible
- Maintain household units
- Connect greywater to conveyance systems where applicable
- Support reuse and recharge
- Organise recharge where feasible

7. Operation & Maintenance (O&M)

- Household level: install screens, clean drains regularly, remove grit/plastic/paper and maintain hygienic conditions.
- Community level: operate treatment units as scheduled; clean filter beds, tanks and screens; maintain pumps and other electro-mechanical devices.
- Check civil structures for leakages and carry out corrective actions.
- Test treated effluent parameters such as BOD, COD, pH, TSS, total nitrogen and total phosphorus.
- Conveyance: de-silt drains, repair leaks/broken sections and address recurring blockages.
- Create grievance mechanisms and maintain service/repair contacts.



Operation and Maintenance

5.1 Introduction

The operation and maintenance of greywater treatment units is a multi-faceted responsibility. It includes in its coverage the conveyance systems, the community-level treatment units, grievance redressal mechanisms and record maintenance protocol. The construction contract for any treatment facility should include operation and maintenance for five years as a mandatory condition.

The responsibilities for Operation and maintenance of various components of greywater management are presented in the table below:

Table 6: Operation and Maintenance Responsibility for Various Greywater Management Components

Sr No	Components of GWM	O&M responsibility
1	HH-level treatment units	HHs through their own funds
2	Group-level treatment units and conveyance systems	HHs concerned collectively
3	Community/GP-level treatment units	GPs through 15th FC funds, MGNREGS, business models/CSR
4	Community/GP-level conveyance systems	GPs through 15th FC funds, MGNREGS, business models/CSR

The operation and maintenance needs for various systems for greywater management are elaborated in the following sections.

8. Funding, Reuse and Financial Management

Up to 5,000 population

GWM financial support: up to ₹280 per capita

Above 5,000 population

GWM financial support: up to ₹660 per capita

IEC & Capacity Building

Up to 5% of total funding for programmatic components

- 30% of the stated village-level amount is to be borne by GPs from 15th Finance Commission grants.
- Each village can utilise a total of ₹1 lakh based on requirements for solid waste and greywater management.
- States may use flexi funds, as per applicable Ministry of Finance guidelines, for local innovations and technology options.
- Reuse and recharge of treated greywater should be promoted for suitable non-potable uses such as irrigation.
- Business models, cost-sharing, cost recovery and revenue-generation approaches can support greywater management.

9. Monitoring and Key Indicators

- GP level: action plan prepared; households covered; treatment-unit connections; reuse of water; absence of stagnant water.
- Block/District level: capacity building, fund use, GP coverage, community treatment arrangements and number of blocks with greywater management.
- Overall GWM: coverage and absence of stagnant water.
 - Greywater treatment: facilities, appropriate technology and O&M.
 - Reuse and recharge: amount of treated greywater reused/recharged.
 - Funding, IEC/community mobilisation, capacity building and O&M.
- Effective monitoring measures implementation quality, progress and outputs.

- ➊ Maintenance activities undertaken for conveyance systems and treatment units
- ➋ Effective convergence of government schemes
- ➌ Efforts towards private partnerships and revenue generation models
- ➍ Implementation of IEC activities as per the plan

6.4 Key Indicators of Monitoring Greywater Management Activities

Key indicators of monitoring greywater management activities are as follows:

No	Component	Key monitoring indicators
1	Overall GWM	▶ Number of GPs with GWM arrangements ▶ Absence of stagnant water in public places ▶ Number of blocks fully covered by GWM arrangements ▶ Details of the GWM arrangements
2	Reduce generation of greywater	▶ Awareness campaigns and activities for reducing the use of fresh water
3	Greywater treatment	▶ Number of HHs, institutions and public places with GWM facilities ▶ Application of HH-level treatment wherever possible ▶ Use of appropriate technology for GWM ▶ Details of O&M of the treatment units
4	Reuse and recharge of greywater	▶ Amount of treated greywater reused and recharged ▶ Number and status of IEC activities conducted for reuse and recharge
5	Funding	▶ Fund disbursement under SBM (G) Phase-II ▶ Use of funds disbursed under SBM (G) Phase-II ▶ Effective convergence of government schemes ▶ Activities for private partnerships ▶ Efforts for revenue generation models ▶ Amount of by-products generated, product-wise amounts sold, details of revenue generated
6.	IEC and community mobilisation	▶ Number of IEC activities conducted ▶ Number of beneficiaries ▶ Number of villages covered
7.	Capacity building	▶ Number of capacity building activities conducted ▶ Type of activities conducted ▶ Number of beneficiaries ▶ Number of villages covered ▶ Number of complaints registered
8.	O&M	▶ Number of complaints addressed

Source: A toolkit for district-level officials on Greywater Management, June 2021.